CHAPTER 7

Relation of the microflora of oral cavity and gallbladder

Mugdha Tendulkar[1] and Reshma Tendulkar[2]
[1]K. J. Somaiya Medical College & Research Centre, Sion, India
[2]Department of Pharmaceutical Chemistry, Vivekanand Education Society's College of Pharmacy, Chembur, India

Introduction

Gallbladder diseases have always been widely spread, socially significant, and most importantly neglected disease for the past numerous decades (Stinton & Shaffer, 2012). The major roadblock in the treatment of any possible disease is the delayed diagnosis of the disease. The diagnosis of the disease upon the appearance of symptoms causes a significant delay till the symptoms appear; the disease has already been spread to such an extent that it has already damaged/infected your human body and now you have only noticed it when it is disrupting your daily activities. At this particular stage, the medical system is only left with the option of controlling the disease or preventing it from worsening rather than curing it. These consequences can be solely attributed to the delayed diagnosis of the disease. In this scenario, what holds importance is the prognosis of the disease or finding an efficient biomarker for the diagnosis of the disease, thus helping us to improve our health status. Unfortunately, Allopathic medicine has restricted boundaries when it comes to treating gallbladder diseases. They only offer surgical methods for treating gallstones; the chemical drugs that are administered have adverse effects on the human body, thus further worsening the scenario (Lammert et al., 2016). Furthermore, the surgical removal methods fail to ensure no recurrence, which means that they can reappear in your lifetime. This complicates the way of treating patients with gallstones. In a nutshell, the overall existing methods of tackling this condition and the responses observed from the patients invariably instigate us to question the existing medicinal paradigm of treating it.

The statistics for the prevalence of gallstones are quite astonishing asserting the diagnosis of about 0.5 million patients in the United States

Gallstone Formation, Diagnosis, Treatment, and Prevention
DOI: https://doi.org/10.1016/B978-0-443-16098-1.00011-4
© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(Stinton et al., 2010). As per various studies, it has been observed that about 80% of all gallstones are left undetected owing to the delayed appearance of symptoms which is the characteristic problem faced in the case of gallstones (Sakorafas et al., 2007). These statistics highlight the alarming rate at which the concern of gallstones is increasing globally.

Our human body constantly strives to be in homeostasis, where organ systems, microbiomes, and metabolism coexist in a specific predefined way. Any glitch in the process destroys the related cascading process which ultimately disrupts the homeostasis of the body. The body regulates these processes with the help of strict checkpoints at certain steps of these processes. An example of the human body's coevolution and coexistence is the composition of microbiomes. Humans, though multicellular eukaryotes, have to constantly interact with other unicellular and multicellular organisms. They have to establish a communal relationship with the microbial community present in the surroundings. This microbial community has a chief role in regulating the health and anatomy of the body. With the synergistic association of the microbial community with the body, we become a holobiont or a superorganism (Sender et al., 2016). This society of microbes residing in our body is assigned the name "microbiome." Nobel prize laureate Joshua Lederberg coined the term "microbiome" to highlight the ecological, communal importance of these microbes which have been residing in our body but have been neglected as precursors of ailments (Lederberg & Mccray, 2001). In the past decades, novel development in the field of genomic technologies has been invented like next-generation sequencing (NGS) and various other tools that have helped the field of science to understand the correlation between the microbiome and the body (Turnbaugh et al., 2007). It has been an increasing interest to evaluate a particular ailment in terms of its change in the existing composition of the microbiome present in the body. It entails the idea of diagnosis of the disease with the help of such potential biomarkers. In most diseases, particularly gallstones, it becomes impossible to identify the disease with the help of blood tests, and in this case, it necessitates the presence of a potential biomarker that will help in the prognosis of the disease, thereby improving the health standards of the patients (Rajab et al., 2020).

Oral microbiome

Oral microbiome refers to the microbial residents in the oral cavity of the human body, which shelters about 700 strains of bacteria (Kilian et al., 2016). These bacteria populate the enamel of the teeth and the soft tissues

of mucosal epithelium of the oral cavity (Deo & Deshmukh, 2019). Oral microbiome is the second largest microbiome in the body, after the gut microbiome (Willis & Gabaldón, 2020). It is observed that the disturbance in the oral microbiome due to constantly changing lifestyles has fatal effects on the overall health of the body (Gao et al., 2018). With the advent of novel technologies, it is possible to untangle the deep associations of oral microbiome with the health of the human body, this leads to clarifying our perception regarding the microbiome (Cho et al., 2021). It has been observed through various studies that these microbes have coevolved with us through millions of years and their composition is completely attributed to our evolution (Davenport et al., 2017). There is a ceaseless effort of the microbiome to be in continual ecological valance with the changing conditions of the human body, which makes their relationship with the body dynamic (Gilbert et al., 2018). It is heavily affected by the numerous aspects of modern lifestyles such as diet and stress, which alters the composition of the oral microbiome and these changing conditions lead the microbiome to arrive at a stage when it is no longer in equilibrium with the body (Sedghi et al., 2021). In order to effectively diagnose a disease, we need to concentrate our attention on the microbial community and its linked parts instead of treating the human body as a whole (Zhang et al., 2015).

Fig. 7.1 illustrates the constructive effects of the oral microbiome on human body.

The oral cavity is the most highly sheltered part of the human body (Mark Welch et al., 2020) The oral cavity provides a characteristic habitat for the innumerable microbial residents which is an important association between their community and the general health of the body (Faran Ali & Tanwir, 2012).

Oral microbiome: a complex ecological community

It is observed that the microbiome not only consists of mere unicellular organisms having a mutual relationship with each other and sharing a common habitat but also forms highly modulated, functionally, and structurally assembled populations habituated as biofilms on the surfaces; they also comprise interspecies association and antagonisms that remarkably confer ecological stability (Ursell et al., 2012). Inside a biofilm, this bacterial microbiome can interact with each other by synthesizing, perceiving, and reacting with diffusible, small molecules (Huang et al., 2011).

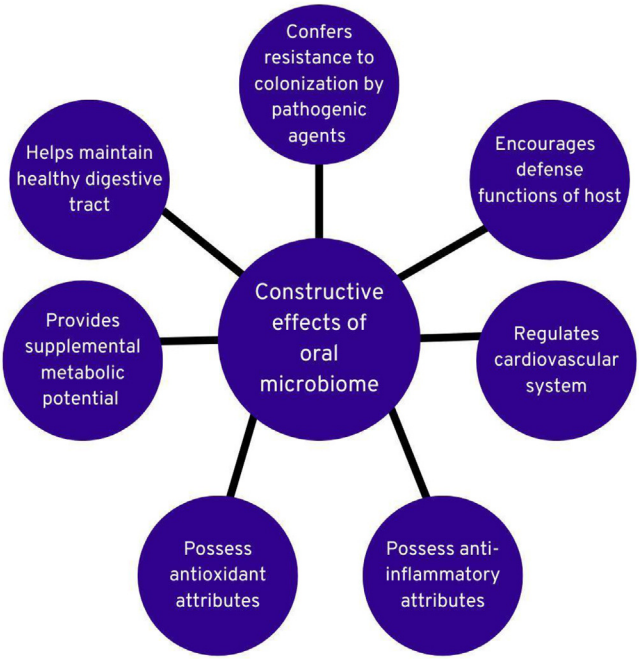

Figure 7.1 Constructive effects of the oral microbiome on human body. This figure explains the beneficial effects conferred by the oral microbiome.

This mechanism of interaction is known as quorum sensing, which extends assistance for competitive defense, host colonization, adaptation to environmental changes, and the formation of biofilm (Li & Tian, 2012). These quorum-sensing processes occurring in the biofilm are also engaged in controlling the pathogenic potential of the bacteria, which plays a notable role in recognizing and controlling bacterial infections (Rutherford & Bassler, 2012). These enable the microbial residents to become more tolerant to the host's defense mechanisms and to the antimicrobial agents.

Oral microbiomes are in charge of the body's numerous crucial immunological, physiological, and metabolic processes (Lee et al., 2021).

Fig. 7.2 portrays the various influences of oral microbiome on the homeostasis of the body.

Any kind of disturbance in the composition of oral microbiome has pronounced effects on health. This constitution of the oral microbiome exhibits great heterogeneity among the numerous areas of the human body, as well as within and among people (Lloyd-Price et al., 2016). It is

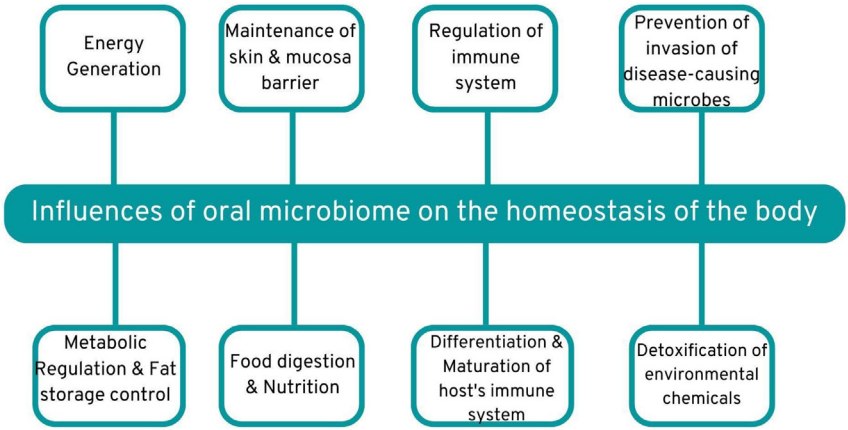

Figure 7.2 Various influences of oral microbiome on the homeostasis of the body. This figure encompasses the varied influences of oral microbiota on the human body.

reported that the microbial community reading in the gut and the skin are distinct enough to give rise to a microbial fingerprint that is considered unique to every person (Wilkins et al., 2017). Regardless of the variations observed in its composition, the cumulative effects of the microbiome among the population are relatively similar.

The mouth is not considered a homogenous habitat for the oral microbiota; on the contrary, it offers several habitats to the bacteria residing in it (Caselli et al., 2020). It includes tongue, cheek, hard and soft palate, teeth, attached gingiva, and attached gingiva. These different habitats exhibit their own unique individual characteristics and hence support the remarkably different bacterial communities (Shade et al., 2012). The humid and moist environment of the oral cavity further supports the growth of microbes and the host also provides nutrients to them such as gingival crevicular fluid (GCF), saliva proteins, and glycoproteins (Marsh & Percival, 2006). The teeth, particularly enamel are the sole non-shedding surfaces of the human body and give rise to extensive opportunities for the growth of the microbiota (Palmer, 2014). The other non-shedding surfaces that further support the existence of the microbiome are the crown, bridgework, implants, and various dental restorations performed by the populations (Rismanchian et al., 2012). These activities promote the formation of biofilm and hence help the microbial residents to thrive in the oral cavity.

More than 700 prokaryotes have been identified from oral microbiomes, most of which can be isolated from the conventional culture techniques (Zhang et al., 2018). Among the identified bacterial species, the statistics as per the extensive literature survey suggest that about 54% are scientifically nomenclatured, 14% are unnamed but are cultivated in the laboratory, and 32% of them are uncultivated species.

Association between oral microbiome and overall health

The composition of the oral microbiome has an enormous influence on the general health of an individual. Any changes in its composition are indicative of an ailment in the human body. There exists a complicated equilibrium exists among the microbial residents in oral cavity (Cho & Blaser, 2012). They can either be accountable for the healthy state known as symbiosis, or a diseased state known as dysbiosis. Dysbiotic microbiome is a community in which the composition of its microbial residents is unbalanced, their diversity is harmed due to ailment or any other reasons (Carding, 2015). As mentioned earlier, the association of oral microbiome with human body is dynamic, which suggests that overall constitution of oral microbiome gets stabilized after childhood, and it remains relatively of the same composition. Taking into consideration the other biological changes occurring during the life of the population, the dynamic equilibrium adjusts itself, and homeostasis is achieved (Bostanci, 2021). These biological changes can be explained as hormonal changes in pregnancy and puberty, the age; the oral microbiome adjusts itself to these changes without causing any harm to its overall health or to its microbial composition (Lamont, 2018). During some specific occurrences, this microbiome might get severely damaged leading to regression of the biofilm, and loss of community present in it; this indicates the presence of a disease or disorder in the human body (Lasserre, 2018). The occurrences contributing to the disruption of the oral microbiome include dysfunction of the salivary gland, which includes changes in the flow of saliva and its composition: diet, smoking, poor oral health, and gingival inflammation(Marcotte & Lavoie, 1998).

Fig. 7.3 depicts the factors leading to dysbiosis.

It is scientifically proven that the bacteria that were initially considered pathogens are present in low numbers at healthy sites (Gao et al., 2018). Most oral diseases occur due to the loss of particular bacterial species from the stable oral microbiome, indicating a disruption in their balance, which

Figure 7.3 Factors leading to Dysbiosis. This figure explains the various factors giving rise to the event of Dysbiosis.

signifies an infection in the body because of an external pathogen (Gao et al., 2018). In dysbiosis, the concentration of the pathogenic microbes reaches a significant amount, which was meager in the stable oral microbiome composition (Flemer et al., 2018).

Bacterial communities in the oral microbiome

For the past decades, numerous studies have been carried out that verify the association of the role of microbiome with the numerous compartments of gastrointestinal tract. This strong correlation portrays its direct influence on the occurrence of gallstone diseases (Gad Elhak et al., 2004). The biliary microbiome heavily corresponds with salivary microbiota. It is also observed that biliary microbiota possess a high resemblance with duodenum microbiome. This shows us strong links between various parts of the human body and their intense links with each other. A disruption in any of the microbiomes denotes malfunctions in the linked microbiomes of the other parts of the body. The oral pathogenic microbial residents

residing heavily influences the gallbladder motility and also control genetic expression of the mucin genes (Muc3, MUC1, MUC4) through the immunomodulatory mechanisms (Figura et al., 1998). This proves the fact that they are the main promoters of gallstone formations. The existence of *Heliobacter pylori* infection in the body significantly aids the gallstone genesis and also increases complications in other gallstone diseases. These complications include pancreatitis, cholangitis, and acute cholecystitis (Fremont-Rahl et al., 2013).

Intestinal bacteria such as *Bifidobacterium, Bacteroides, Peptostreptococcus, Clostridium, Escherichia coli*, and *Eubacterium*—which are known to take part in epimerization and oxidation of the biliary acids—disturb the enterohepatic circulation, and are a major factor promoting gallstone genesis (Bostanci, 2021). As per medical science, the only permanent way of getting rid of gallbladder diseases is cholecystectomy, which is the surgical removal of the gallbladder present in the human body (Deo & Deshmukh, 2019). This leads to major compositional changes in the microbiomes of the various parts of the body especially the microbiomes associated with the gastrointestinal tract. This results in a high risk of developing colorectal and stomach malignancies in the future. A need of further research is obviously required to ascertain the probability of using the meticulous analysis and compositions of the microbiomes as a potential biomarker for the prognosis of various diseases associated with the gastrointestinal tract (Attaallah et al., 2013).

The oral cavity is the main window of the human body through which continuous interactions between the external environment and the internal systems take place. It is also an entrance for the respiratory and digestive tracts in the human body, as a consequence of this important role, it is highly vascularized. This intense vascularization makes the oral cavity possible to hint at the occurrence of various systemic diseases like cardiovascular (Farshad et al., 2004), rheumatoid arthritis (Dyck et al., 2003), endocrine (Faran Ali & Tanwir, 2012), and cancers (Lee et al., 2021) by compositional changes in the microbiomes of the oral cavity. Various studies have been successfully carried out that prove the deep connection of the various diseases with the compositional changes in the oral microbiome. This is an important fact that can be effectively exploited to cure several diseases with the help of the prognosis of the disease, leading to effective medical treatment at the right time, thus saving the patient's life and improving the standards of the medical system (Lasserre, 2018).

The total observed surface area of the oral cavity is approximately $214.7 \pm 12.9\ cm^2$ (Kasprzak et al., 2015). Oral microbiota is one of the most unique and comprehensive ecosystems comprising bacteria, fungi, viruses, protozoans, and archaea. There exist more than 700 prokaryotic species that interact with each other to produce extensive biofilms (Verma et al., 2018). An individual's microbiome is highly specific at the species level, though it exhibits some geographical differences overall. It is also observed that the number of various microbes can fluctuate depending upon the changing lifestyles and other factors. Till now, 50–200 microbial residents belonging to 15 genera residing in the human microbiome have been isolated, which are known to be present in almost every individual (Casarin et al., 2013; Zhang et al., 2015). A healthy microbiome predominantly consists of these six prominent types:

1. 96% of the microbial residents are associated with Firmicutes (genus Streptococcus, family Veillonellaceae, genus Granulicatella)
2. Actinobacteria (genus Corynebacterium, Rothia, Actinomyces)
3. Proteobacteria (genus Neisseria, Haemophilus)
4. Bacteroidetes (genus Prevotella, Capnocytophaga, Porphyromonas)
5. Fusobacteria (genus Fusobacterium) (Palmer, 2014)
6. Spirochaetes (Rudi et al., 1999)
7. The remaining 4% of the total microbial residents belong to Chlamydia, SR1Euryarchaeota, Griniastos Chloroflexi, Synergistetes, TM7, and Tenericutes.

There exists a ceaseless flow of microorganisms from the external environment apart from these local microbial residents that are already present in the oral cavity. It is also known that almost all the microbes belonging to the genera Alphaproteobacteria are the transients of the environment, and those belonging to the genera Bacteroidetes are linked with the host (Shade et al., 2012; Sherafat et al., 2012). The composition of microbiota is attributed to change by fluctuating environmental situations, medications, age, contracted disorders, etc (Lasserre, 2018; Lederberg & Mccray, 2001).

The research findings proposed by Shen et al. showed that, of the 13 novel biliary bacteria isolated with the help of whole-metagenome shotgun sequencing (WMS), 8 belonged to the human oral microbiome taxa and the remaining 5 belonged to the taxa of environmental origin (Mark Welch et al., 2020). The oral bacterial residents of the oral cavity can directly modulate the microbiomes present in the gallbladder and those of the upper gastrointestinal tract; these factors promote gallstone synthesis

and the development of other gallstone diseases as well. The oral bacterial species are known to disrupt the synthesis of nitric acid (NO), which is an essential cofactor of the enzyme eNOS. This enzyme is present in both the colon and the vascular network, this reduces the expression of the antioxidant protein Nrf2. These incidents also reduce the bioavailability of nitric oxide, which leads to a significant rise in the concentration of reactive oxygen species in the human body.

Oral bacterial residents have been extensively involved in the pathogenesis of gallstones. Although there is sheer lack of clear understanding of its role in the genesis of gallstones among people. According to the United Nations population study, the Third National Health and Nutritional Examination Survey 1988–94 (NHANES III) included 995 adults suffering from gallbladder diseases. The study also had 10,232 controls belonging to the age group of 20–74 years. The findings of this univariate study analysis suggested that the predictors of gallbladder diseases are missing teeth (OR = 4.8, 95% CI 3.1–7.4, $P < .001$) and poor oral hygiene (odds ratio (OR) = 1.7, 95% confidence interval (CI) 1.1–1.25, $P = .02$). The analysis verified that missing teeth are an individualistic precursor of gallbladder diseases modified (OR = 1.7, 95% CI 1.1–2.8, $P = .02$) (Lamont, 2018). Analysis of biliary fluids in patients suffering from gallbladder diseases reported that the most prevalent residents in the gastrointestinal tract are Proteobacteria, Fusobacteria, Bacteroidetes, Firmicutes, TM7, Synergistetes, and Actinobacteria (Zhang et al., 2015; Hassler et al., 2022). These microbial residents of genus Pyramidobacter belonging to phylum Synergistetes have been predominantly isolated and derived from the oral cavity. The majority of the bacterial residents found in the bile of patients were 3 genera of Enterobacteriaceae (Klebsiella, Escherichia, and an unclassified genus) and Pyramidobacter. The leading genera present in the bile of patients suffering from gallbladder diseases like Haemophilus, Prevotella, and Rothia were comparatively more equivalent with genera residing in the oral microbiota (Takahashi et al., 2014). Another factor playing an important role in the etiopathogenesis of gallbladder diseases is oxidative stress. Patients suffering from gallbladder diseases have elevated levels of oxidative stress in the mucosal epithelium of the gallbladder; this leads to altered gallbladder absorption and secretion of biliary constituents like glycoproteins and mucins (Bulajic et al., 2002; Davenport et al., 2017; Huang et al., 2011). These events of supersaturation of biliary fluids significantly aid in gallstone genesis.

The important genetic factor controlling the development of gallstones is transcription nuclear factor-erythroid (NF-E) 2-related factor 2 (Nrf2) which acts as a transcription regulator of a wide panel of genes associated with detoxification and drug metabolism and as an antioxidant in the human body (Aleksunes & Manautou, 2007). During *in vivo* studies on mice, it is observed that the protein expression of Transcription factor (*Nrf2*) is also found to be remarkably increased in gallstone-resistant AKR/J strain of mice as against gallstone-susceptible C57L/J strain mice. It is genetically reported that the Transcription factor (*Nrf2*) maps to the Lith 1 loci on the chromosomes. This identifies Transcription factor (*Nrf2*) as a putative *Lith 1* gallstone gene (Willis & Gabaldón, 2020; Jiménez-Hernández et al., 2019).

Studies confirm the position of the salivary microbiome as a potential diagnostic biomarker for cardiovascular (Xu et al., 2018) and metabolic disorders, which are very closely associated with the occurrence of gallbladder diseases (Carding, 2015; Jiménez-Hernández et al., 2019). The authentication of observed bacterial residents residing in the oral cavity through the quantitative polymerase chain reaction (PCR) presented results showing the healthy controls occupied notably decreased levels of the microbial resident Granulicatella adiacens ($P = .023$). It was also observed that the healthy controls had a high ratio of Peptococcus to Granulicatella ($P < .05$) as opposed to that observed in the controls studied in the metabolic syndrome (Marsh & Percival, 2006; Mendez-Sanchez et al., 2001). The researchers stress the idea of employing these bacterial residents as a diagnostic tool for the prognosis of systemic disorders. According to a research study performed by Teles et al., they examined the correlation between the levels of systemic biomarkers and oral parameters of inflammation. The crux of their research investigation was the quality and quantity of host responses to the oral bacterial community were more fundamental to systemic atherothrombotic coronary occurrences. It is also proven that the remarkably elevated risk of atherosclerosis is directly linked with increased levels of fibrinogen, plasma C-reactive protein (CRP), and proinflammatory cytokine levels. The results hint at the ability of the oral bacterial residents as a potential biomarker for systemic diseases especially periodontal pathogens in atherogenesis (Li & Tian, 2012). Another research investigation verified the existence of 23 oral synergistic bacterial residents, either residing independently or mutually in the atherosclerotic plaques in the individuals who went through endarterectomy. Out of 23 bacterial residents, 5 of them

(*Prevotella nigrescens, Porphyromonas gingivalis, Campylobacter rectus, Porphyromonas endodontalis*, and *Prevotella intermedia*) are distinctive to coronary plaques (Grigor'eva et al., 2011; Hänninen, 1991). As a consequence of these extensive research studies, the relationship between atherosclerosis, metabolic syndromes, and gallbladder diseases is well established (Sherafat et al., 2012; Shreiner et al., 2015; Zaura et al., 2009).

The congenital immunostimulation by infective microflora of oral cavity remarkably influences the release of cholecystokinin, which is a prominent contributor responsible for filing and emptying of gallbladder (Xu et al., 2018; Zaura et al., 2009). The oral microbiota modifies genetic expression of mucin genes (Muc3, MUC1, & MUC4) by immunomodulatory mechanisms. This affects the mucin gel accretion; a nucleation matrix responsible for pathogenesis of the cholesterol gallstones (Wade, 2013; Walker et al., 2018). The profuse participation of oral microflora in the formation of gallstones has notably promoted research for employing these oral microbial residents as a primary and secondary treatment of gallstones (Goodson et al., 2009; Hänninen, 1991).

Helicobacter pylori and *Helicobacter* spp. and gallstones

Innumerable research investigations have proved the direct influence of *H. pylori* on gallstone genesis (Casadevall & Pirofski, 2000; Farrell et al., 2012; Fremont-Rahl et al., 2013; Shen et al., 2015; Corrêa et al., 2019; Sharma et al., 2018; Shen et al., 2015; Wang et al., 2021; Willis & Gabaldón, 2020; Sipos et al., 2001). *H. pylori* is a Gram-negative, motile and spiral-shaped bacterium. After the identification of *H. pylori*, numerous have been isolated from the gastrointestinal tract, especially from the stomach and intestinal compartments of numerous species of mammals. Approximately 18–25 species of *Helicobacter* spp. have been identified now (Cho et al., 2021).

Certain specific species of Heliobacter which include *H. bilis* and *H. hepaticus* reside in intestinal tracts and occupy biliary ducts and liver (Peters et al., 2017; Rudi et al., 1999; Verma et al., 2018). Due to the hypersensitivity of *H. pylori* to bile, it was unable to be detected in the bile samples of patients or in the mucosal epithelium of the gallbladder in patients suffering from biliary tract disorders. It was also exhibited that the bile salts that were unconjugated were more toxic than conjugated salts (Dyck et al., 2003).

The bacterial species *Helicobacter* spp. and *H. pylori* present in the gallstones, biliary samples, and gallbladder tissues were identified by numerous analytical techniques like cultures (Di Stefano et al., 2022; Flemer et al., 2018; Gao et al., 2018), urease test, immunohistochemistry (Doron & Gorbach, 2008), immunohistochemical stain (Chhibber-Goel et al., 2016), by 16S rRNA PCR-amplification, by PCR-DGGE (Tajeddin et al., 2016), by nested and multiplex PCR (Abayli et al., 2005; Bulajic et al., 2002; Le Bars et al., 2017; Mark Welch et al., 2020; Mendez-Sanchez et al., 2001; Turnbaugh et al., 2007; Verma et al., 2018), by ELISA following a preabsorption step using *H. pylori* antigens (Mendez-Sanchez et al., 2001), both by PCR & by an ELISA study of the particular antigen (*H. pylori* stool antigen), by a Western blot analysis of Helicobacter antibodies (Hamada et al., 2009), WMS sequencing, 16S rRNA sequencing on biliary samples and Giemsa (Stinton et al., 2010). The prominent reason for conflicting results is the use of different methods of detection of *H. pylori* in the experimental samples by different researchers.

According to research investigation, analysis of the mucosal epithelium of the gallbladder after cholecystectomy in 94 patients suffering from symptomatic gallbladder diseases was considered. These mucosal epithelia were analyzed in terms of *H. pylori* with the help of immunohistochemical stain, Giemsa, and urease test. This study manifests the existence of *H. pylori* in mucosal gallbladder epithelium as observed in 37% (based on data obtained from any of those three tests conducted) of patients suffering from gallbladder diseases (Attaallah et al., 2013). Kountouras et al. conducted a study in Greece that exhibited positive *H. pylori* serology in 51.3% of patients suffering from bile stones and pancreatic disorders (Kountouras et al., 2014). In addition, a cross-reactivity between *H. hepaticus*, *H. pylori*, and *H. bilis* has been observed (Nilsson et al., 2005).

To identify Heliobacter in the gallstones of patients suffering from gallstone and biliary diseases, a research excavation was carried out in Iran. The bile samples and gallstones procured from 33 Iranian patients were made to undergo the following techniques such as culture, multiplex PCR, and rapid urease test (Stathopoulos et al., 2006). The results indicated that in 18.1% of gallstones and from 12.1% of biliary samples from these 33 patients, the DNA of *H. pylori* was detected using Multiplex PCR. On the other hand, negative results for all samples were obtained for rapid urease and culture tests. PCR findings also turned out to be negative in the control group (Shreiner et al., 2015).

Another study analyzed 102 biliary samples from patients suffering from biliary disorders which included 74 patients suffering from gallbladder diseases in order to detect *Heliobacter* spp. in the cultures. The results obtained indicated that the *Heliobacter* spp. Did not show any growth in the cultures whereas the DNA of *H. pylori* was detected in 3.92% of the samples by PCR (Sherafat et al., 2012). The findings also suggested that there is no remarkable association identified between the presence of bacteria in the cultures and the development of the diseases.

As per an Iranian case-control study, 77 individuals with gallstones (n = 52) and without gallstones (n = 25) were considered as subjects for the study (Fatemi et al., 2018). All the biliary culture samples were negative for the presence of Heliobacter species. *H. pylori* is extremely arduous to develop on the culture media owing to its microaerophilic attributes (Farshad et al., 2004). As compared with the routine bacterial culture, it was also found that the PCR technique has high specificity and high sensitivity while analyzing cultures.

According to another study, the DNA of *Heliobacter* spp. was identified in the biliary samples of 19.2% of individuals suffering from gallstones (Fatemi et al., 2018). Out of this, eight (15.3%) were designated to be *H. pylori* and two (3.9%) were found to the DNA of *H. bilis*. There were no Heliobacter species observed in gallstones when detected by PCR. The results also suggested a relation between the existence of DNA of *H. pylori* in biliary samples and the development of gallstones. Statistically speaking, there is no such relation between the development of gallbladder diseases and the three enterohepatic *Heliobacter* spp. which are *H. hepaticus, H. pullorum*, and *H. bilis* (Hamada et al., 2009). On the other hand, 41% of patients suffering from gallbladder diseases had a notably high ($P = .029$) occurrence of *H. hepaticus* infection in the biliary samples as against the samples collected from patients suffering from other disorders. The researchers deduced that *H. hepaticus* is highly interconnected with liver and bile disorders in the human population (Cai & Chen, 2014). Often, *H. hepaticus* antibodies were identified in individuals having gallbladder diseases.

A clinical study in Brazil of biliary tissues and gallbladders collected from 46 subjects with gallbladder diseases and 18 without suffering from these diseases were considered. The study exhibited a direct and individualistic relationship between the existence of Heliobacter DNA (based on the findings detected by nested PCR assay) and the development of gallbladder diseases in the tissues of gallbladder ($P = .009$; OR = 14.72; 95%

CI = 1.97 to 108.90). The genetic experiments displayed that the genomic sequences of the 16s rRNA were found to be more than 99% analogous to that of *H. pylori* (Silva et al., 2003). These findings from these experiments supported the proposed hypothesis that the presence of Heliobacter DNA is associated with the genesis of gallbladder disorders in humans.

The urease produced by the Heliobacter species was observed to stimulate the calcium precipitation, and invariably promote the formation of gallstones in the human body. The infection of *H. pylori* significantly influences the physiology of gallbladder stone genesis and its further impediments which also include biliary cancer, cholangitis, cholecystitis, and pancreatitis. The mechanism involved is the release of vasoactive and proinflammatory substances like IL-6, tumor necrosis factor (TNF)-α, and IL-1 present in large amounts (Kasprzak et al., 2015). These substances are associated with the genesis of gallbladder diseases and inflammatory disorders of the gallbladder (Takahashi et al., 2014). In addition, other factors that stimulate gallstone formation include oxidative stress and free radical reactions in the wall of the gallbladder, and the biliary fluid.

It has been found experimentally that the CagA protein present in *H. pylori* has homology with aminopeptidase, which remarkably increases the risk of gallstone genesis. It is also reported that *H. pylori* also stimulates the genesis of gallstones by behaving as an alien body in order to create a nidus, where there is a high risk of development of gallstones in the area around it, or it may synthesize nucleating proteins like immunoglobins or hydrolyzing enzymes. There are several contrasting studies as well that question the existing findings available to us. A study proposed by Griniastsos et al. reported that *H. pylori* fails to possess any lithogenic attribute for the pathogenesis of gallstones. Another study by Stathopoulos et al. presented their study suggesting that the infection of *H. pylori* significantly affects the functioning of gallbladder consequently causing a decrease in its accumulation potential (Stathopoulos et al., 2006).

Based on the retroactive research study conducted on 7803 subjects in Beijing exhibited that the occurrence of gallstones in group of individuals having *H. pylori* (+) was found to be 1.53 times reduced as compared to the group of individuals exhibiting *H. pylori* (−) ($P = .012$) (Xu et al., 2018). A research study based in Mexico revealed that there is no association between the development of gallstones and the Heliobacter infection in gallbladder (Mendez-Sanchez et al., 2001).

The mere existence of *H. pylori* infection heavily influences the incidence of complications in gallbladder diseases which involve pancreatitis, chronic and acute cholecystitis, and cholangitis. According to an Iranian research study, a correlation was observed between the existence of DNA of acute cholecystitis and *H. pylori*. In addition to this, a recent meta-analysis confirmed the direct correlation between the elevated risk of acute cholecystitis and *H. pylori* infection OR = 3.022 (95% CI, 1.897−4.815; I2 = 20.1%) (Cen et al., 2018). Furthermore, the odds ratio (95% CI) for research in Asian and non-Asian areas of the world were slightly different and was found to be 3.75 (1.83−7.71) and 2.25 (1.29−3.89) correspondingly. These findings reveal that there might be demographic and racial distinctions in the etiology of gallstones.

After cholecystectomy, it was reported that the extent of colonization of *H. pylori* in the mucosal epithelium of gallbladder notably decreased from 69.6% to 41.3% of individuals ($P < .0001$) (Gad Elhak et al., 2004). Research studies based in China and Japan verified that *H. pylori* infection is directly correlated with the occurrence of gallstones. Furthermore, it was asserted that the annihilation of *H. pylori* leads to the avertion of the occurrence of gallstones in the human body (Binda et al., 2022).

Besides, a human research study proposed 20 years ago by Figura et al. revealed that *H. pylori* infection in biliary specimens is a potential element behind the elevated risk of gallstone genesis (Figura et al., 1998). The primary contribution of Heliobacter infection in development of gallstones is still unclear to medical sciences. More research studies are required to highlight the role played by *Heliobacter* spp. in the etiopathogenesis of gallstones.

Gallstone microbiome

Through powerful, bacteriological cultivations, electron microscopy and molecular genetic techniques, the existence of alive bacteria identified in gallstones has been manifested. Peng et al. researched gallstone microbiome and biliary fluid in individuals suffering from gallstones. The results suggested that 30% of the cultured samples procured from gallstones secreted phospholipase A2 and β-glucuronidase. Out of the total samples, 14 genera of the bacterial residents were reported to be present in the cholesterol gallstones, and 8 genera of bacteria were reported in the biliary samples (Dewhirst et al., 2010). The predominant bacterial resident in the microbiomes of gallstones and bile was *Pseudomonas* spp. It was researched

that the strains of *Pseudomonas aeruginosa* possessed elevated levels of phospholipase A2 and had the highest activity of β-glucuronidase. Another alternative mechanism for the genesis of pigment gallstones is biofilms formation (Stewart et al., 1987). The dominant factor that promotes aggregation in this process is the glycocalyx which is basically an anionic glycoprotein.

The distinctive differences observed in practical metagenomes of these microbiomes in the cholesterol and pigmented gallstones were unveiled through numerous research studies (Di Stefano et al., 2022). Furthermore, the genes responsible for the formation of biofilm were predominantly isolated from the species *Enterococcus* and *Klebsiella* in the pigment gallstones. In addition, bile resistance genes were also found namely *Serratia, Bacillus, Escherichia, Klebsiella*, and *Shigella* extracted from the cholesterol gallstones. Most of the gallstones analyzed that Gram(+) bacterial residents were absent in pigment gallstones. On the other hand, large amount of genes associated with the metabolism of carbohydrates were exhibited in the cholesterol gallstones. It was also reported that the genes involving protein metabolism were found to be more active in these gallstones.

Conclusions

In a nutshell, based on the current evidence, there is considerable interest in the employment of oral microbiomes to assess the risk of developing gallstones, which requires further validation in a large population. The advancement of diagnostic biomarkers can be successfully exploited for predicting the severity of the disease and also in ascertaining the treatment extended, determining the need to perform a biopsy, therefore, aiding in the prognosis of the disease. In the near future, a medical system with constantly improving paradigms and patients will be given the treatment not as per their diagnosed disease but as per their personalized molecular signatures. The prognosis of the disease significantly aids in its treatment as the disease under consideration is mostly asymptomatic. Thus, more extensive research work has to be performed to unravel the transmission patterns from the oral cavity to the gallbladder and the interaction between them. There is a need to research the mechanisms through which this microbiota interacts and how they alter their composition upon the changes in the human body. This can be a breakthrough in identifying novel prognostic approaches by modulating the oral microbial communities through the oral and gallbladder bacterial transmission intervention.

References

Abayli, B., Colakoglu, S., Serin, M., Erdogan, S., Isiksal, Y. F., Tuncer, I., Koksal, F., & Demiryurek, H. (2005). Helicobacter pylori in the aetiology of cholesterol gallstones. *Journal of Clinical Gastroenterology*, *39*, 134–137.

Attaallah, W., Yener, N., Ugurlu, M. U., Manukyan, M., Asmaz, E., & Aktan, A. O. (2013). Gallstones and Concomitant Gastric Helicobacter pylori Infection. *Gastroenterology Research and Practice*, *2013*, 643109. Available from https://doi.org/10.1155/2013/643109.

Binda, C., Gibiino, G., Coluccio, C., Sbrancia, M., Dajti, E., Sinagra, E., Capurso, G., Sambri, V., Cucchetti, A., Ercolani, G., & Fabbri, C. (2022). Biliary diseases from the microbiome perspective: How microorganisms could change the approach to benign and malignant diseases. *Microorganisms*, *10*(2), 312. Available from https://doi.org/10.3390/microorganisms10020312.

Bostanci, N., et al. (2021). Dysbiosis of the human oral microbiome during the menstrual cycle and vulnerability to the external exposures of smoking and dietary sugar. *Frontiers in Cellular and Infection Microbiology*. Available from https://doi.org/10.3389/fcimb.2021.625229.

Bulajic, M., Maisonneuve, P., Schneider-Brachert, W., Müller, P., Reischl, U., Stimec, B., Lehn, N., Lowenfels, A. B., & Löhr, M. (2002). Helicobacter pylori and the risk of benign and malignant biliary tract disease. *Cancer*, *95*, 1946–1953. Available from https://doi.org/10.1002/cncr.10893.

Cai, J. S., & Chen, J. H. (2014). The mechanism of enterohepatic circulation in the formation of gallstone disease. *Journal of Membrane Biology*, *247*, 1067–1082. Available from https://doi.org/10.1007/s00232-014-9715-3.

Carding, S., et al. (2015). Dysbiosis of the gut microbiota in disease. *Microbial Ecology in Health and Disease*. Available from https://doi.org/10.3402/mehd.v26.26191.

Casadevall, A., & Pirofski, L. A. (2000). Host-pathogen interactions: basic concepts of microbial commensalism, colonization, infection, and disease. *Infection and Immunity*, *68*(12), 6511–6518. Available from https://doi.org/10.1128/IAI.68.12.6511-6518.2000.

Casarin, R. C. V., Barbagallo, A., Meulman, T., Santos, V. R., Sallum, E. A., Nociti, F. H., Duarte, P. M., Casati, M. Z., & Gonçalves, R. B. (2013). Subgingival biodiversity in subjects with uncontrolled type-2 diabetes and chronic periodontitis. *Journal of Periodontal Research*, *48*, 30–36. Available from https://doi.org/10.1111/j.1600-0765.2012.01498.x.

Caselli, E., Fabbri, C., D'Accolti, M., Soffritti, I., Bassi, C., Mazzacane, S., & Franchi, M. (2020). Defining the oral microbiome by whole-genome sequencing and resistome analysis: The complexity of the healthy picture. *BMC Microbiology*, *20*(1). Available from https://doi.org/10.1186/s12866-020-01801-y.

Cen, L., Pan, J., Zhou, B., Yu, C., Li, Y., Chen, W., & Shen, Z. (2018). HelicobacterPylori infection of the gallbladder and the risk of chronic cholecystitis and GSD: A systematic review and meta-analysis. *Helicobacter*, *23*. Available from https://doi.org/10.1111/hel.12457.

Chhibber-Goel, J., Singhal, V., Bhowmik, D., Vivek, R., Parakh, N., Bhargava, B., & Sharma, A. (2016). Linkages between oral commensal bacteria and atherosclerotic plaques in coronary artery disease patients. *NPJ Biofilms and Microbiomes*, *2*, 7. Available from https://doi.org/10.1038/s41522-016-0009-7.

Cho, I., & Blaser, M. J. (2012). The human microbiome: at the interface of health and disease. *Nature Reviews Genetics*, *13*, 260–270.

Cho, Y.-D., Kim, K.-H., Lee, Y.-M., Ku, Y., & Seol, Y.-J. (2021). Oral microbiome and host health: Review on current advances in genome-wide analysis. *Applied Sciences*, *11*(9), 4050. Available from https://doi.org/10.3390/app11094050.

Corrêa, J. D., Fernandes, G. R., Calderaro, D. C., Mendonça, S. M. S., Silva, J. M., Albiero, M. L., Cunha, F. Q., Xiao, E., Ferreira, G. A., Teixeira, A. L., Mukherjee, C., Leys, E. J., Silva, T. A., & Graves, D. T. (2019). Oral microbial dysbiosis linked to worsened periodontal condition in rheumatoid arthritis patients. *Scientific Reports*, *9*(1), 8379. Available from https://doi.org/10.1038/s41598-019-44674-6.

Davenport, E. R., Sanders, J. G., Song, S. J., Amato, K. R., Clark, A. G., & Knight, R. (2017). The human microbiome in evolution. *BMC Biology*, *15*(1). Available from https://doi.org/10.1186/s12915-017-0454-7.

Deo, P. N., & Deshmukh, R. (2019). Oral microbiome: Unveiling the fundamentals. *Journal of Oral and Maxillofacial Pathology*, *23*(1), 122–128. Available from https://doi.org/10.4103/jomfp.JOMFP_304_18.

Dewhirst, F. E., Chen, T., Izard, J., Paster, B. J., Tanner, A. C., Yu, W. H., Lakshmanan, A., & Wade, W. G. (2010). The human oral microbiome. *Journal of Bacteriology*, *192*, 5002–5017. Available from https://doi.org/10.1128/JB.00542-1.

Di Stefano, M., Polizzi, A., Santonocito, S., Romano, A., Lombardi, T., & Isola, G. (2022). Impact of oral microbiome in periodontal health and periodontitis: A critical review on prevention and treatment. *International Journal of Molecular Sciences*, *23*(9), 5142. Available from https://doi.org/10.3390/ijms23095142.

Doron, S., & Gorbach, S. L. (2008). Bacterial infections: Overview. *International Encyclopedia of Public Health*, 273–282. Available from https://doi.org/10.1016/B978-012373960-5.00596-7.

Dyck, P. A., Hoda, F., Osmer, E. S., & Green, R. M. (2003). Microarray analysis of hepatic gene expression in gallstone-susceptible and gallstone-resistant mice. *Mammalian Genome*, *14*, 601–610. Available from https://doi.org/10.1007/s00335-003-2269-9.

Faran Ali, S. M., & Tanwir, F. (2012). Oral microbial habitat a dynamic entity. *Journal of Oral Biology and Craniofacial Research*, *2*(3), 181–187. Available from https://doi.org/10.1016/j.jobcr.2012.07.001.

Farrell, J. J., Zhang, L., Zhou, H., Chia, D., Elashoff, D., Akin, D., Paster, B. J., Joshipura, K., & Wong, D. T. (2012). Variations of oral microbiota are associated with pancreatic diseases including pancreatic cancer. *Gut*, *61*, 582–588. Available from https://doi.org/10.1136/gutjnl-2011-300784.

Farshad, S. H., Alborzi, A., Malek Hosseini, S. A., Oboodi, B., Rasouli, M., Japoni, A., & Nasiri, J. (2004). Identification of helicobacter pylori DNA in Iranian patients with gallstones. *Epidemiology and Infection*, *132*, 1185–1189. Available from https://doi.org/10.1017/S0950268804002985.

Fatemi, S. M., Doosti, A., Shokri, D., Ghorbani-Dalini, S., Molazadeh, M., Tavakoli, H., Minakari, M., & Tavakkoli, H. (2018). Is there a correlation between helicobacter pylori and enterohepatic helicobacter species and gallstone cholecystitis? *Middle East Journal of Digestive Diseases*, *10*, 24–30. Available from https://doi.org/10.15171/mejdd.2017.86.

Figura, N., Cetta, F., Angelico, M., Montalto, G., Cetta, D., Pacenti, L., Vindigni, C., Vaira, D., Festuccia, F., De Santis, A., Rattan, G., Giannace, R., Campagna, S., & Gennari, C. (1998). Most helicobacter pylori-infected patients have specific antibodies, and some also have H. pylori antigens and genomic material in bile: Is it a risk factor for gallstone formation? *Digestive Diseases and Sciences*, *43*(4), 854–862. Available from https://doi.org/10.1023/a:1018838719590.

Flemer, B., Warren, R. D., Barrett, M. P., Cisek, K., Das, A., Jeffery, I. B., Hurley, E., O'Riordain, M., Shanahan, F., & O'Toole, P. W. (2018). The oral microbiota in colorectal cancer is distinctive and predictive. *Gut*, *67*, 1454–1463. Available from https://doi.org/10.1136/gutjnl-2017-314814.

Fremont-Rahl, J. J., Ge, Z., Umana, C., Whary, M. T., Taylor, N. S., Muthupalani, S., Carey, M. C., Fox, J. G., & Maurer, K. J. (2013). An analysis of the role of the indigenous microbiota in cholesterol gallstone pathogenesis. *PLoS ONE*, *8*e70657. Available from https://doi.org/10.1371/journal.pone.0070657.

Gad Elhak, N., Abd Elwahab, M., Nasif, W. A., Abo-Elenein, A., Abdalla, T., el-Shobary, M., Haleem, M., Yaseen, A., el-Ghawalby, N., & Ezzat, F. (2004). Prevalence of helicobacter pylori, gastric myoelectrical activity, gastric mucosal changes and dyspeptic symptoms before and after laparoscopic cholecystectomy. *Hepatogastroenterology*, *51*, 485–490.

Gao, L., Xu, T., Huang, G., Jiang, S., Gu, Y., & Chen, F. (2018). Oral microbiomes: more and more importance in oral cavity and whole body. *Protein Cell*, *9*(5), 488–500. Available from https://doi.org/10.1007/s13238-018-0548-1.

Gao, L., Xu, T., Huang, G., Jiang, S., Gu, Y., & Chen, F. (2018). Oral microbiomes: More and more importance in oral cavity and whole body. *Protein and Cell*, *9*(5), 488–500. Available from https://doi.org/10.1007/s13238-018-0548-1.

Gilbert, J. A., Blaser, M. J., Caporaso, J. G., Jansson, J. K., Lynch, S. V., & Knight, R. (2018). Current understanding of the human microbiome. *Nature Medicine*, *24*(4), 392–400. Available from https://doi.org/10.1038/nm.4517.

Goodson, J. M., Groppo, D., Halem, S., & Carpino, E. (2009). Is obesity an oral bacterial disease? *J. Dent. Res.*, *88*, 519–523. Available from https://doi.org/10.1177/0022034509338353.

Grigor'eva, I. N., Logvinenko, E. V., Yamlikhanova, A. Y., & Romanova, T. I. (2011). View of cholelithiasis through the prism of metabolic syndrome. *Bull. Sib. Branch RAMS.*, *31*, 72–78.

Hamada, T., Yokota, K., Ayada, K., Hirai, K., Kamada, T., Haruma, K., Chayama, K., & Oguma, K. (2009). Detection of Helicobacter hepaticus in human bile samples of patients with biliary disease. *Helicobacter*, *14*, 545–551. Available from https://doi.org/10.1111/j.1523-5378.2009.00729.x.

Hänninen, M. L. (1991). Sensitivity of Helicobacter pylori to different bile salts. *European Journal of Clinical Microbiology & Infectious Diseases*, *10*, 515–518. Available from https://doi.org/10.1007/BF01963941.

Hassler, K. R., Collins, J. T., Philip, K., & Jones, M. W. (2022). *Laparoscopic cholecystectomy.StatPearls [Internet]*. Island, FL: StatPearls: Treasure.

Huang, R., Li, M., & Gregory, R. L. (2011). Bacterial interactions in dental biofilm. *Virulence*, *2*(5), 435–444. Available from https://doi.org/10.4161/viru.2.5.16140.

Jiménez-Hernández, N., Serrano-Villar, S., Domingo, A., Pons, X., Artacho, A., Estrada, V., Moya, A., & Gosalbes, M. J. (2019). Modulation of saliva microbiota through prebiotic intervention in HIV-infected individuals. *Nutrients*, *11*(6), 1346. Available from https://doi.org/10.3390/nu11061346.

Kasprzak, A., Szmyt, M., Malkowski, W., Przybyszewska, W., Helak-Lapaj, C., Seraszek-Jaros, A., Surdacka, A., Małkowska-Lanzafame, A., & Kaczmarek, E. (2015). Analysis of immunohistochemical expression of proinflammatory cytokines (IL-1alpha, IL-6, and TNF-alpha) in gallbladder mucosa: Comparative study in acute and chronic calculous cholecystitis. *Folia Morphologica*, *74*, 65–72. Available from https://doi.org/10.5603/FM.2015.0011.

Kilian, M., Chapple, I. L. C., Hannig, M., Marsh, P. D., Meuric, V., Pedersen, A. M. L., Tonetti, M. S., Wade, W. G., & Zaura, E. (2016). The oral microbiome—An update for oral healthcare professionals. *British Dental Journal*, *221*(10), 657–666. Available from https://doi.org/10.1038/sj.bdj.2016.865.

Kountouras, J., Tsiaousi, E., Trigonis, S., Zavos, C., & Kouklakis, G. (2014). Helicobacter pylori infection in a Greek cohort with biliary disease. *British Journal of Biomedical*

Science, *71*, 178–179. Available from https://doi.org/10.1080/09674845.2014.11669984.

Lammert, F., Gurusamy, K., Ko, C. W., Miquel, J. F., Méndez-Sánchez, N., Portincasa, P., Van Erpecum, K. J., Van Laarhoven, C. J., & Wang, D. Q. H. (2016). Gallstones. *Nature Reviews Disease Primers*, *2*. Available from https://doi.org/10.1038/nrdp.2016.24.

Lamont, R. J., et al. (2018). The oral microbiota: Dynamic communities and host interactions. *Nature Reviews Microbiology*, *16*(12), 745–759. Available from https://doi.org/10.1038/s41579-018-0089-x.

Lasserre, J. F., et al. (2018). Oral microbes, biofilms and their role in periodontal and peri-implant diseases. *Materials (Basel)*, *11*(10). Available from https://doi.org/10.3390/ma11101802.

Le Bars, P., Matamoros, S., Montassier, E., Le Vacon, F., Potel, G., Soueidan, A., Jordana, F., & de La Cochetière, M. F. (2017). The oral cavity microbiota: Between health, oral disease, and cancers of the aerodigestive tract. *Canadian Journal of Microbiology*, *63*, 475–492. Available from https://doi.org/10.1139/cjm-2016-0603.

Lederberg, J., & Mccray, A. (2001). Ome sweet 'omics—A genealogical treasury of words. *Scientist*, *15*, 8–10.

Lee, Y. H., Chung, S. W., Auh, Q. S., Hong, S. J., Lee, Y. A., Jung, J., Lee, G. J., Park, H. J., Shin, S. I., & Hong, J. Y. (2021). Progress in oral microbiome related to oral and systemic diseases: An update. *Diagnostics*, *11*(7). Available from https://doi.org/10.3390/diagnostics11071283.

Li, Y. H., & Tian, X. (2012). Quorum sensing and bacterial social interactions in biofilms. *Sensors*, *12*(3), 2519–2538. Available from https://doi.org/10.3390/s120302519.

Lloyd-Price, J., Abu-Ali, G., & Huttenhower, C. (2016). The healthy human microbiome. *Genome Medicine*, *8*(1). Available from https://doi.org/10.1186/s13073-016-0307-y.

Marcotte, H., & Lavoie, M. C. (1998). Oral microbial ecology and the role of salivary immunoglobulin A. *Microbiology and Molecular Biology Reviews*, *62*(1), 71–109. Available from https://doi.org/10.1128/MMBR.62.1.71-109.1998.

Mark Welch, J. L., Ramírez-Puebla, S. T., & Borisy, G. G. (2020). Oral microbiome geography: Micron-scale habitat and niche. *Cell Host and Microbe*, *28*(2), 160–168. Available from https://doi.org/10.1016/j.chom.2020.07.009.

Marsh, P. D., & Percival, R. S. (2006). The oral microflora—Friend or foe? Can we decide? *International Dental Journal*, *56*(4), 233–239. Available from https://doi.org/10.1111/j.1875-595x.2006.tb00107.x, FDI World Dental Press Ltd.

Mendez-Sanchez, N., Pichardo, R., González, J., Sánchez, H., Moreno, M., Barquera, F., Estevez, H. O., & Uribe, M. (2001). Lack of association between Helicobacter sp. colonization and gallstone disease. *Journal of Clinical Gastroenterology*, *32*, 138–141. Available from https://doi.org/10.1097/00004836-200102000-00009.

Nilsson, I., Shabo, I., Svanvik, J., & Monstein, H. J. (2005). Multiple displacement amplification of isolated DNA from human gallstones: Molecular identification of Helicobacter DNA by means of 16S rDNA-based pyrosequencing analysis. *Helicobacter*, *10*, 592–600. Available from https://doi.org/10.1111/j.1523-5378.2005.00361.x.

Palmer, R. J. (2014). Composition and development of oral bacterial communities. *Periodontology 2000*, *64*(1), 20–39. Available from https://doi.org/10.1111/j.1600-0757.2012.00453.x.

Peters, B. A., Wu, J., Pei, Z., Yang, L., Purdue, M. P., Freedman, N. D., Jacobs, E. J., Gapstur, S. M., Hayes, R. B., & Ahn, J. (2017). Oral microbiome composition reflects prospective risk for esophageal cancers. *Cancer Research*, 77, 6777–6787. Available from https://doi.org/10.1158/0008-5472.CAN-17-1296.

Rajab, I. M., Majerczyk, D., Olson, M. E., Addams, J. M. B., Choe, M. L., Nelson, M. S., & Potempa, L. A. (2020). C-reactive protein in gallbladder diseases: Diagnostic and therapeutic insights. *Biophysics Reports*, *6*(2–3), 49–67. Available from https://doi.org/10.1007/s41048-020-00108-9.

Rismanchian, M., Babashahi, A., Goroohi, H., Shahabouee, M., Yaghini, J., & Badrian, H. (2012). Microflora around teeth and dental implants. *Dental Research Journal*, *9*(2), 215. Available from https://doi.org/10.4103/1735-3327.95239.

Rudi, J., Rudy, A., Maiwald, M., & Stremmel, W. (1999). Helicobacter sp. are not detectable in bile from German patients with biliary disease. *Gastroenterology*, *116*, 1016–1017. Available from https://doi.org/10.1016/S0016-5085(99)70096-X.

Rutherford, S. T., & Bassler, B. L. (2012). Bacterial quorum sensing: Its role in virulence and possibilities for its control. *Cold Spring Harbor Perspectives in Medicine*, *2*(11). Available from https://doi.org/10.1101/cshperspect.a012427.

Sakorafas, G. H., Milingos, D., & Peros, G. (2007). Asymptomatic cholelithiasis: Is cholecystectomy really needed? A critical reappraisal 15 years after the introduction of laparoscopic cholecystectomy. *Digestive Diseases and Sciences*, *52*(5), 1313–1325. Available from https://doi.org/10.1007/s10620-006-9107-3.

Sedghi, L., DiMassa, V., Harrington, A., Lynch, S. V., & Kapila, Y. L. (2021). The oral microbiome: Role of key organisms and complex networks in oral health and disease. *Periodontology 2000*, *87*(1), 107–131. Available from https://doi.org/10.1111/prd.12393.

Sender, R., Fuchs, S., & Milo, R. (2016). Are we really vastly outnumbered? Revisiting the ratio of bacterial to host cells in humans. *Cell*, *164*(3), 337–340. Available from https://doi.org/10.1016/j.cell.2016.01.013.

Shade, A., Peter, H., Allison, S. D., Baho, D. L., Berga, M., Bürgmann, H., Huber, D. H., Langenheder, S., Lennon, J. T., Martiny, J. B. H., Matulich, K. L., Schmidt, T. M., & Handelsman, J. (2012). Fundamentals of microbial community resistance and resilience. *Frontiers in Microbiology*, *3*. Available from https://doi.org/10.3389/fmicb.2012.00417.

Sharma, N., Bhatia, S., Sodhi, A. S., & Batra, N. (2018). Oral microbiome and health. *AIMS Microbiology*, *4*(1), 42–66. Available from https://doi.org/10.3934/microbiol.2018.1.42.

Shen, H., Ye, F., Xie, L., Yang, J., Li, Z., Xu, P., Meng, F., Li, L., Chen, Y., Bo, X., Ni, M., & Zhang, X. (2015). Metagenomic sequencing of bile from gallstone patients to identify different microbial community patterns and novel biliary bacteria. *Scientific Report*, *5*, 17450. Available from https://doi.org/10.1038/srep17450.

Shen, H., Ye, F., Xie, L., Yang, J., Li, Z., Xu, P., Meng, F., Li, L., Chen, Y., Bo, X., Ni, M., & Zhang, X. (2015). Metagenomic sequencing of bile from gallstone patients to identify different microbial community patterns and novel biliary bacteria. *Scientific Reports*, *5*, 17450. Available from https://doi.org/10.1038/srep17450.

Sherafat, S. J., Tajeddin, E., Majidi, M. R. S., Vaziri, F., Alebouyen, M., Alizade, A. H. M., Mojarad, E. N., & Zali, M. R. (2012). Lack of association between Helicobacter pylori infection and biliary tract diseases. *Polish Journal of Microbiology*, *61*, 319–322. Available from https://doi.org/10.33073/pjm-2012-044.

Shreiner, A. B., Kao, J. Y., & Young, V. B. (2015). The gut microbiome in health and in disease. *Current Opinion in Gastroenterology*, *31*(1), 69–75. Available from https://doi.org/10.1097/MOG.0000000000000139.

Silva, C. P., Pereira-Lima, J. C., Oliveira, A. G., Guerra, J. B., Marques, D. L., Sarmanho, L., Cabral, M. M., & Queiroz, D. M. (2003). Association of the presence of Helicobacter in gallbladder tissue with GSD and cholecystitis. *Journal of Clinical Microbiology*, *41*, 5615–5618. Available from https://doi.org/10.1128/JCM.41.12.5615-5618.2003.

Sipos P., Krisztina H., Blázovics A., Fehér J. Cholecystitis, gallstones and free radical reactions in human gallbladder. Medical Science Monitor 2001;7:84–88.

Stathopoulos, P., Zundt, B., Spelsberg, F. W., Kolligs, L., Diebold, J., Goke, B., & Jüngst, D. (2006). Relation of gallbladder function and Helicobacter pylori infection to gastric mucosa inflammation in patients with symptomatic cholecystolithiasis. *Digestion*, *73*, 69–74. Available from https://doi.org/10.1159/000092746.

Stewart, L., Smith, A. L., Pellegrini, C. A., Motson, R. W., & Way, L. W. (1987). Pigment gallstones form as a composite of bacterial microcolonies and pigment solids. *Annals of Surgery*, *206*(3), 242–251. Available from https://doi.org/10.1097/00000658-198709000-00002.

Stinton, L. M., & Shaffer, E. A. (2012). Epidemiology of gallbladder disease: Cholelithiasis and cancer. *Gut and Liver*, *6*(2), 172–187. Available from https://doi.org/10.5009/gnl.2012.6.2.172.

Stinton, L. M., Myers, R. P., & Shaffer, E. A. (2010). Epidemiology of gallstones. *Gastroenterology Clinics of North America*, *39*(2), 157–169. Available from https://doi.org/10.1016/j.gtc.2010.02.003.

Takahashi, Y., Yamamichi, N., Shimamoto, T., Mochizuki, S., Fujishiro, M., Takeuchi, C., Sakaguchi, Y., Niimi, K., Ono, S., Kodashima, S., Mitsushima, T., & Koike, K. (2014). Helicobacter pylori infection is positively associated with gallstones: A large-scale cross-sectional study in Japan. *Journal of Gastroenterology*, *49*(5), 882–889. Available from https://doi.org/10.1007/s00535-013-0832-z.

Turnbaugh, P. J., Ley, R. E., Hamady, M., Fraser-Liggett, C. M., Knight, R., & Gordon, J. I. (2007). The human microbiome project. *Nature*, *449*(7164), 804–810. Available from https://doi.org/10.1038/nature06244.

Ursell, L. K., Metcalf, J. L., Parfrey, L. W., & Knight, R. (2012). Defining the human microbiome. *Nutrition Reviews*, *70*(1), S38–S44. Available from https://doi.org/10.1111/j.1753-4887.2012.00493.x.

Verma, D., Garg, P. K., & Dubey, A. K. (2018). Insights into the human oral microbiome. *Archives of Microbiology*, *200*, 525–540. Available from https://doi.org/10.1007/s00203-018-1505-3.

Verma, D., Garg, P. K., & Dubey, A. K. (2018). Insights into the human oral microbiome. *Archives of Microbiology*, *200*, 525–540. Available from https://doi.org/10.1007/s00203-018-1505-3.

Wade, W. G. (2013). The oral microbiome in health and disease. *Pharmacological Research*, *69*, 137–143. Available from https://doi.org/10.1016/j.phrs.2012.11.006.

Walker, M. Y., Pratap, S., Southerland, J. H., Farmer-Dixon, C. M., Lakshmyya, K., & Gangula, P. R. (2018). Role of oral and gut microbiome in nitric oxide-mediated colon motility. *Nitric Oxide*, *73*, 81–88. Available from https://doi/10.1016/j.niox.2017.06.003.

Wang, L., Chen, J., Jiang, W., Cen, L., Pan, J., Yu, C., Li, Y., Chen, W., Chen, C., & Shen, Z. (2021). The relationship between Helicobacter pylori infection of the gallbladder and chronic cholecystitis and cholelithiasis: A systematic review and meta-analysis. *Canadian Journal of Gastroenterology and Hepatology*, *2021*, 8886085. Available from https://doi.org/10.1155/2021/8886085.

Wilkins, D., Leung, M. H. Y., & Lee, P. K. H. (2017). Microbiota fingerprints lose individually identifying features over time. *Microbiome*, *5*(1). Available from https://doi.org/10.1186/s40168-016-0209-7.

Willis, J. R., & Gabaldón, T. (2020). The human oral microbiome in health and disease: From sequences to ecosystems. *Microorganisms*, *8*(2). Available from https://doi.org/10.3390/microorganisms8020308.

Willis, J. R., & Gabaldón, T. (2020). The human oral microbiome in health and disease: From sequences to ecosystems. *Microorganisms*, *8*, 308. Available from https://doi.org/10.3390/microorganisms8020308.

Xu, M. Y., Ma, J. H., Yuan, B. S., Yin, J., Liu, L., & Lu, Q. B. (2018). Association between Helicobacter pylori infection and gallbladder diseases: A retrospective study. *Gastroenterology and Hepatology*, *33*, 1207–1212. Available from https://doi.org/10.1111/jgh.14054.

Zaura, E., Keijser, B. J., Huse, S. M., & Crielaard, W. (2009). Defining the healthy "core microbiome" of oral microbial communities. *BMC Microbiology*, *9*, 259. Available from https://doi.org/10.1186/1471-2180-9-259.

Zhang, Y. J., Li, S., Gan, R. Y., Zhou, T., Xu, D. P., & Li, H. B. (2015). Impacts of gut bacteria on human health and diseases. *International Journal of Molecular Sciences*, *16*(4), 7493–7519. Available from https://doi.org/10.3390/ijms16047493.

Zhang, Y., Wang, X., Li, H., Ni, C., Du, Z., & Yan, F. (2018). Human oral microbiota and its modulation for oral health. *Biomedicine and Pharmacotherapy*, *99*, 883–893. Available from https://doi.org/10.1016/j.biopha.2018.01.146.

Zhang, F. M., Yu, C. H., Chen, H. T., Shen, Z., Hu, F. L., Yuan, X. P., & Xu, G. Q. (2015). Helicobacter pylori infection is associated with gallstones: Epidemiological survey in China. *World Journal of Gastroenterology*, *21*, 8912–8919. Available from https://doi.org/10.3748/wjg.v21.i29.8912.